WEEKS 1-2

Internship - spring semester

Σαντοριναίου Άννα

DOCKER

- Advantage of working with docker
- What a docker image is and how to build one
- How to build a dockerfile
- biribiri

ROS2

- Advantages
- Nodes & how they communicate through topics, services, actions
- How to launch nodes
- turtlesim & rqt console examples
- Package creation
- tf2

URDF

OR

how to specify the geometry & organization of a robot

- A Tutorial on how to build a visual robot model and also:
  - Apply movable joints
  - Visual and collision elements
  - Add physical properties to the model, such as mass
  - Publish the state of the robot as a tf2 message
  - View the simulation

ROS2 & controll

- Acts as a bridge, between hardware and control algorithms
- Robot hardware

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  Hardware components

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  Resource manager

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  Controller manager:

  Joint State cont. Position cont. Velocity cont.

PID controller

- Can be used directly, by sending references through topics (or in a chain)
- command interface, for a specific DoF or an actuator
- state interface
- examples

MPC

- Basic procedure
- Advantages: constraints application, and more
- Works well for non-linear systems
- Usually uses Linear Parameter Varying model

- Principles of MPC:

1. internal dynamic model
2. cost function over horizon
3. optimization algorithm

Control Barrier Functions

- Initial goal: safety of the system
- Invariance can be seen as an example of a safety property
- Nagumo's theorem
- Barrier functions are added to cost functions to avoid undesirable regions
- Barrier certificates:
  - unsafe set Cu
  - set of initial conditions Co
  - a function B with a certain definition

Implementations

- Soft Surgical Robots & Tissue Contact Safety
- Cutting task in endoscopic submucosal dissection
- Mediates medical treatments to bound the estimated internal state within a prescribed safe regime

- Dynamic Walking on Stepping Stones
- Automotive Systems: Automatic Cruise Control and Lane Keeping

Thank you